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Machine Learning-Based Domain Risk Assessment for Cybersecurity Monitoring in Industrial Systems

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Abstract

In the field of cybersecurity, malicious website classification plays a crucial role in protecting industrial systems. For this reason, research has been undertaken to analyze cybersecurity threats, with the long-term objective of developing methods for the effective detection and classification of malicious websites. This article evaluates the use of domain features for classifying malicious websites with machine learning methods. The feature vector consisted of 44 infrastructural, lexical, structural, and reputation-related characteristics. The model comparison included Logistic Regression, Support Vector Machine, Random Forest, AdaBoost, and XGBoost. Experiments were conducted on 247,730 URLs from the malicious_phish dataset (2021), with features extracted as part of this study in April 2026. The most predictive features were related to domain registration history, DNS infrastructure, and reputation-based rankings, as confirmed by ANOVA F-test, SHAP values, XGBoost gain, and permutation importance. Validation of the best-performing model on 1000 active phishing domains from the PhishDestroy list dated 30 May 2026, achieved a recall of 70%, while the application of a three-tier risk scale allowed 84.9% of domains to be flagged as malicious or suspicious.

Keywords: cybersecurity; machine learning; phishing; data-driven monitoring task; website classification; industrial systems

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1. Introduction

As Internet services continue to expand, domains used for criminal purposes remain one of the key elements of cyberattack infrastructure. According to the ENISA Threat Landscape 2025 report, phishing remains the dominant initial intrusion vector, accounting for approximately 60% of observed cases [1]. A particular challenge for protection systems is the dynamic nature of these threats, resulting from the short life cycle of many malicious domains [2,3].

In cybersecurity, malicious website classification can support the protection of organizational and industrial systems against digital threats. Modern industrial environments increasingly rely on internet-connected infrastructure, including SCADA-related services, industrial IoT devices, remote access mechanisms, and cloud-based process management platforms. As a result, they may be exposed to phishing campaigns, malicious redirects, malware distribution, and command-and-control communication [4,5]. Domain-level threat

detection can therefore be treated as a data-driven monitoring task, in which measurable domain attributes are analyzed to identify potentially high-risk destinations before direct interaction with website content is required.

Traditional blocklists and reputation systems have limited ability to detect newly emerging threats. Many malicious domains remain active only for a short period and may disappear before they are added to public security databases, while evasion techniques used by attackers further complicate their identification [2]. Machine learning enables websites to be classified using measurable domain features, including DNS records, WHOIS data, HTTP responses, lexical properties of domain names, structural properties of URLs, and reputation-related indicators. However, many studies focus on selected feature categories or evaluate models only on a separated part of the same dataset [6–9].

In this study, malicious website detection is considered a supporting cybersecurity monitoring process rather than a direct process-control mechanism. The analyzed URLs or domain names are represented by infrastructural, lexical, structural, and reputation-related attributes and assigned a risk score supporting security decisions. In practical deployment, such a score may be used in DNS resolvers, secure web gateways, proxy servers, e-mail gateways, or monitoring components protecting industrial and organizational networks. Its usefulness depends not only on classification quality, but also on the cost and availability of feature acquisition, especially for network-based attributes such as DNS, WHOIS, and HTTP data.

The objective of this study is to comparatively evaluate five machine learning algorithms for malicious domain detection using 44 infrastructural, lexical, structural, and reputation-related features, and to assess their suitability for domain-level cybersecurity monitoring. Features were extracted in April 2026 from 247,730 URLs in the malicious_phish dataset published in 2021. The compared algorithms were Logistic Regression, Support Vector Machine, Random Forest, AdaBoost, and XGBoost. Additional validation was conducted on 31 manually selected domains and 1000 active phishing URLs from the PhishDestroy list dated 30 May 2026, providing assessment under conditions closer to practical deployment.

The main contributions of this work are summarized as follows:

- A 44-dimensional domain feature vector combining infrastructural, lexical, structural, and reputation-related attributes, enabling systematic comparison of the four feature categories under consistent experimental conditions.
- A cost-based three-tier risk classification framework with decision thresholds optimized for multiple operational cost scenarios, including high-availability production environments and critical infrastructure.
- A cross-method feature importance analysis using the ANOVA F-test, SHAP values, XGBoost gain, and permutation importance, supporting the observed dominance of infrastructural features across the evaluated methods.
- An experimental assessment of temporal validity and classification errors to characterize the robustness and limitations of the proposed approach under operationally realistic conditions.
- An external evaluation on contemporary phishing domains and a latency analysis of the domain assessment pipeline to assess the practical applicability of the method in cybersecurity monitoring workflows.

2. Related Works

Malicious domain detection can be based on several categories of information [10]. Lexical features describe the form of the address or domain name, including its length, character entropy, the presence of digits, special characters, and words suggesting an

attempt to obtain user data. Structural features refer to the construction of the full URL, such as the number of parameters, path length, and the presence of percent encoding. Infrastructural features include DNS configuration, WHOIS data, and information derived from HTTP headers. Reputation-related features use the presence of a domain in popularity or reputation sources, such as Tranco.

Türk et al. [9] presented a comprehensive comparison of machine learning, deep learning, and optimization-supported models on the malicious_phish dataset. The authors reported very high performance for the ELECTRA model, with accuracy and F1-score reaching 99%, and pointed to the growing role of language models in URL analysis. This approach is based mainly on the textual representation of the address, which supports short inference time, but does not include information about the actual domain infrastructure and registration context.

Haraldsdóttir et al. [7] focused on phishing detection using digital certificates and domain name features. Data were collected from PhishTank, Tranco, and Censys, and the feature vector was extended with domain name representations obtained using a BiLSTM model. Enriching certificate-based features with domain-related information increased the F1-score from 0.77 to 0.89, while the time-based evaluation achieved an F1-score of 0.88. Results show that combining infrastructure-related data with domain name properties can improve classification performance.

Mvula et al. [8] investigated the classification of COVID-19-related domain names. Their results showed that a limited set of carefully selected lexical, structural, and reputation-related features can provide high classification performance. The authors emphasized the importance of the number of subdomain levels and Shannon entropy. This study is relevant to the present article because it demonstrates the usefulness of features that can be obtained without analyzing the full website content.

Aljabri et al. [6] analyzed lexical, network-based, and content-based features for malicious URL detection. In their experiment, feature selection was performed using correlation analysis, ANOVA, and the chi-square test. The best performance was achieved by the Naive Bayes classifier, with an accuracy of 96%. The study confirms that combining several categories of attributes can improve detection results, but also requires careful feature selection and control of the cost of obtaining them.

The analyzed studies rarely combine a broad set of infrastructural, lexical, structural, and reputation-related features with additional validation on active phishing domains outside the training dataset. This article extends this direction by using all four feature categories and evaluating the model on an external PhishDestroy sample. A comparison of feature categories used in selected studies is presented in Table 1.

Table 1. Comparison of feature categories used in selected studies.

Paper	Infrastructural	Lexical	Structural	Reputation-Related
Türk et al. (2025) [9]	–	✓	✓	–
Haraldsdóttir et al. (2024) [7]	✓	✓	–	–
Mvula et al. (2022) [8]	–	✓	✓	✓
Aljabri et al. (2022) [6]	✓	✓	–	–
This study	✓	✓	✓	✓

Note: "✓" indicates that a given feature category was included in the study, while "–" indicates that it was not included.

3. Materials and Methods

3.1. URL Dataset

The experiments were carried out using the malicious_phish dataset, published in 2021, which contains 651,191 URLs labeled into four categories: benign (428,103), defacement (96,457), phishing (94,111) and malware (32,520). Since the source data included only URLs and class labels, the domain-level features used in this study were extracted separately. Due to the computational and network costs of feature extraction, 250,000 URLs were sampled using stratified random sampling to preserve the original class proportions. Domain-level duplicates were identified by extracting the registered domain (eTLD + 1), with one representative record retained for each base domain, resulting in 247,730 records. Class labels were taken directly from the published malicious_phish dataset without independent verification.

For binary classification, the benign label was mapped to the safe class (0), whereas the defacement, phishing, and malware labels were merged into the malicious class (1). The resulting class ratio was approximately 2.8:1, indicating moderate imbalance that was taken into account during model training and evaluation. The binary class distribution is presented in Figure 1.

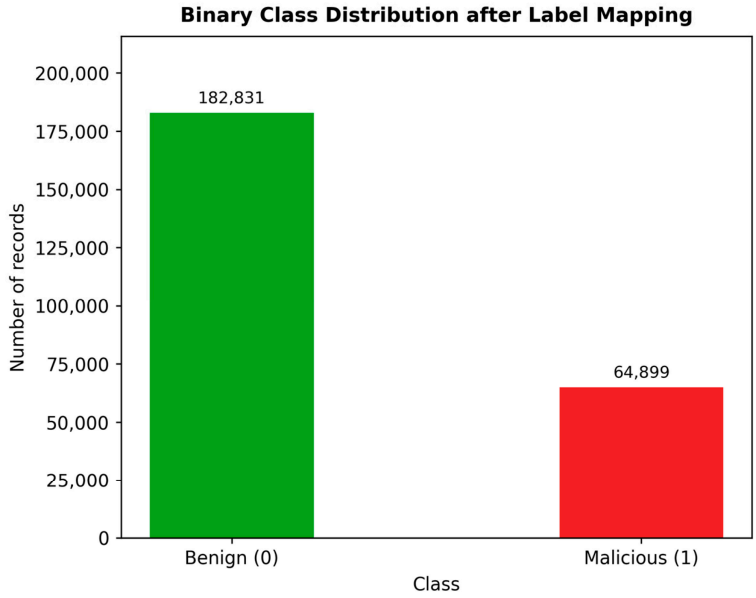


Figure 1. Binary class distribution in the prepared dataset.

3.2. Domain Feature Extraction

For each URL, a feature vector consisting of 44 attributes was constructed from data extracted in April 2026. The features were grouped according to the type of information they represent and the source from which they were obtained. Four main categories were distinguished: infrastructural, lexical, structural, and reputation-related features. The groups of features used in the study are summarized in Table 2.

Infrastructural features described the technical configuration and registration status of the domain. This group included DNS record counts, domain resolution status, SPF, DKIM and DMARC indicators, WHOIS-based attributes such as domain age, expiration time and registrar group, as well as HTTP-based information on the availability of robots.txt, security.txt and Web Application Firewall detection. These attributes represented both network configuration and registration-related properties of the analyzed domain.

Lexical features characterized the form of the domain name itself. This group included domain length, digit ratio, number of hyphens and dots, Shannon entropy, the maximum length of a digit sequence, and indicators of suspicious keywords (a dictionary of 25 terms)

or brand-related terms (a list of 18 brand names). These attributes were computed directly from the domain name and captured patterns such as unusual character composition or excessive use of digits and separators in malicious domains.

Table 2. Feature groups used in the proposed domain representation.

Category	Group	Number of Features	Type
Infrastructural	DNS	12	network-based
	WHOIS	6	network-based
	HTTP	6	network-based
Lexical	Domain name	9	local
Structural	URL	6	local
	Levenshtein	2	local
Reputation-related	Tranco	3	local

Structural features described the full URL. They included URL length, path depth, the number of query parameters, percent encoding, and the number of subdomains. Two additional Levenshtein-based attributes were used to measure similarity between the analyzed domain and a predefined list of 24 protected brand names, covering global payment, e-mail, social media, and e-commerce services, thereby supporting the detection of typosquatting-like patterns.

Reputation-related features were derived from the Tranco ranking [11]. This group included the domain position in the Tranco Top 1M ranking and binary indicators of presence in the Top 1M and Top 10k. For subdomains not directly present in the ranking, the corresponding base domain was also checked, allowing the reputation of the parent domain to be included.

Feature extraction was implemented in Python 3.12.12 (Python Software Foundation, Wilmington, DE, USA) using dedicated libraries for DNS, WHOIS, and HTTP data collection. Network-dependent features were obtained through parallel queries with defined timeouts, while lexical, structural, and reputation-related attributes were computed locally. To improve reproducibility and manage the computational cost of network-based extraction, the process was performed in batches with intermediate results stored in Parquet files. Connection and read timeouts were set to 6 s and 12 s, respectively. In case of timeout or DNS failure, affected features were assigned a value of -1 and subsequently handled during median imputation. Repeated WHOIS queries were memoized using lru_cache, while HTTP responses were cached using requests_cache with a 180-day expiration period.

3.3. Data Splitting and Preprocessing

The prepared dataset was divided into training and test subsets using the GroupShuffleSplit method in a 75:25 ratio. The base domain was used as the grouping variable. This procedure was applied to reduce the risk of domain leakage, which may occur when URLs assigned to the same domain are simultaneously present in the training and test subsets. In the case of a standard random split, such a situation could lead to an overestimation of the model performance [9].

As a result of the split, the training subset contained 188,222 records representing 58,077 unique base domains, whereas the test subset contained 59,508 records representing 19,360 unique base domains. The distribution of benign and malicious records in both subsets is presented in Figure 2.

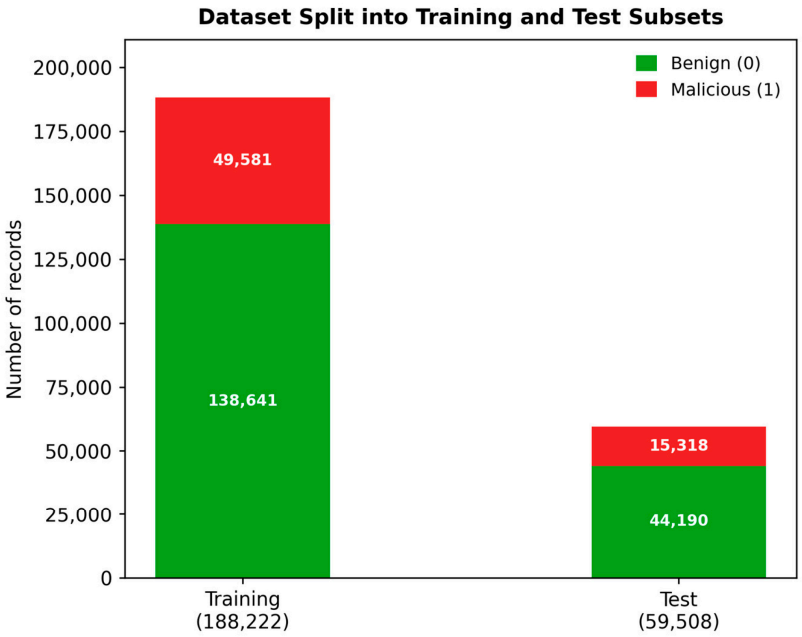


Figure 2. Distribution of safe and malicious records in the training and test subsets.

Preprocessing was performed after dataset splitting and was fitted only on the training subset. Numerical variables were imputed using the median and standardized. Categorical variables were imputed using the most frequent value and transformed by one-hot encoding, while binary variables were imputed using the mode. After categorical encoding, the number of input columns increased from 44 to 57.

Class imbalance was handled separately for each model using only the training data [12]. After the train/test split, imputation was fitted on the training subset and applied to both subsets. For AdaBoost, oversampling was performed using SMOTENC from imbalanced-learn 0.14.1 before one-hot encoding and standardization. The test subset was not oversampled and retained its original class distribution throughout the experiments. The complete order of operations, from raw URL data to final domain classification, is shown in Figure 3.

To assess whether the −1 sentinel values assigned to domain_age_days and days_to_expire when WHOIS data were unavailable introduced an encoding artifact, an ablation study compared three feature representations: the baseline −1 encoding (Variant A), median imputation with binary availability flags retained (Variant B), and median imputation with the availability flags removed (Variant C).

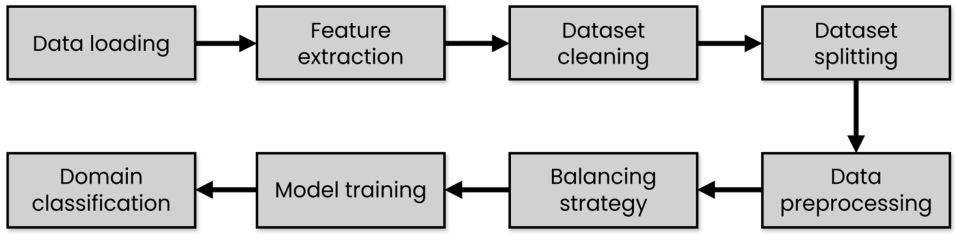


Figure 3. Classification pipeline architecture.

The same preprocessing scheme was used for all classifiers. Class imbalance was handled within the training data only, either by class weighting or by SMOTENC, depending on the model configuration. This allowed the evaluated models to be compared under consistent input conditions.

3.4. Machine Learning Classification Models

In this study, five classification algorithms were compared: Random Forest, XGBoost, AdaBoost, Support Vector Machine (SVM) with an RBF kernel, and Logistic Regression, covering tree-based ensembles, a margin-based classifier, and a linear baseline model. Random Forest combined 600 bagged decision trees with class_weight = 'balanced_subsample' [13], while XGBoost used gradient boosting with built-in regularization, 800 trees, and learning_rate = 0.05 [14]. AdaBoost sequentially trained shallow decision trees as weak estimators, assigning greater weight to previously misclassified observations [15].

SVM was used to model non-linear decision boundaries in the feature space [16,17], with C = 2.0 and gamma = 'scale', whereas Logistic Regression served as a linear baseline classifier trained with the saga solver. Class imbalance was handled using model-specific strategies: Random Forest used class_weight = 'balanced_subsample', SVM and Logistic Regression used class_weight = 'balanced', AdaBoost used SMOTENC on the training data, and XGBoost used scale_pos_weight based on the negative-to-positive class ratio (approximately 2.8). For all algorithms, random_state = 42 was used to ensure experiment reproducibility.

Hyperparameter values were selected on the basis of preliminary experiments on the training subset and configurations commonly used in the literature for this class of problems. For the tree-based ensembles, the number of estimators was increased until validation performance stabilized, while the remaining XGBoost parameters were kept at conservative values that act as additional regularization and support stable, directly comparable behavior of all five models.

3.5. Classification Performance Metrics

The models were evaluated using metrics suitable for imbalanced classification problems. Accuracy, precision, recall, F1-score and Matthews correlation coefficient (MCC) were calculated from the confusion matrix. In addition, ROC AUC and PR AUC were used to assess threshold-independent classification performance. Let TP, TN, FP and FN denote true positives, true negatives, false positives and false negatives, respectively. The confusion-matrix-based metrics were defined by Equations (1)–(5):

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{1}$$

$$Precision = \frac{TP}{TP+FP} \tag{2}$$

$$Recall = \frac{TP}{TP+FN} \tag{3}$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision+Recall} \tag{4}$$

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \tag{5}$$

MCC takes into account all elements of the confusion matrix, which makes it useful for evaluating classifiers on imbalanced datasets [18]. ROC AUC was used to assess the ability of the model to separate the two classes independently of a specific decision threshold. PR AUC was used as a complementary metric because it provides additional information on minority class performance. Feature importance was assessed using the ANOVA F-test, calculated only on the training set.

3.6. Integration with Cybersecurity Monitoring Pipelines

In industrial network environments, domain-level threat detection can be considered a supporting cybersecurity monitoring process. DNS queries, URLs observed in network

traffic, or HTTP connection attempts may provide domain names that can be analyzed as security-related events. A typical pipeline includes: (1) event capture from DNS resolver logs or network traffic; (2) domain-level feature extraction; (3) risk classification by a machine learning model; and (4) alert generation or forwarding to a SIEM system or network access control layer [4,19].

The practical applicability of this process depends not only on classification performance but also on the time required to construct the feature vector. In the proposed method, lexical, structural, Levenshtein-based, and Tranco-related attributes are computed locally and introduce only limited processing overhead. In contrast, DNS, WHOIS, and HTTP attributes require communication with external services and account for most of the end-to-end latency. Their availability and response times may also vary between domains, which can affect the stability of real-time monitoring.

For these reasons, the proposed method is intended primarily for risk assessment and cybersecurity monitoring rather than direct synchronous traffic control. Lightweight pre-filtering methods, such as learned Bloom filter architectures, may support rapid initial screening [20]. Network-based attributes can then be collected asynchronously or cached, while the complete feature vector refines the assessment and generates an additional security signal without unnecessarily delaying legitimate communication.

Let u denote the analysed URL, $\phi(u)$ the corresponding feature vector, and f_θ the trained classifier. The model score can be expressed as:

$$s(u) = f_\theta(\phi(u)) \tag{6}$$

where $s(u) \in [0,1]$ represents the estimated maliciousness score assigned by the model. In the initial binary setting, the threshold $\tau = 0.5$ was used, with domains satisfying $s(u) \geq \tau$ classified as malicious and domains with $s(u) < \tau$ classified as benign. However, the asymmetric consequences of false-positive and false-negative decisions motivated the subsequent introduction of an intermediate-risk category for domains requiring additional automated or manual verification.

3.7. Three-Tier Risk Classification Framework

To address the limitations of a single binary threshold, a three-tier decision function was introduced using two thresholds, τ_1 and τ_2, where $\tau_1 < \tau_2$:

$$d(u) = \begin{cases} \textit{low risk}, & \textit{if } s(u) < \tau_1, \\ \textit{elevated risk}, & \textit{if } \tau_1 \leq s(u) < \tau_2, \\ \textit{high risk}, & \textit{if } s(u) \geq \tau_2. \end{cases} \tag{7}$$

Low-risk domains are processed according to the standard security policy. Elevated-risk domains are forwarded for additional verification using content analysis, sandboxing, threat-intelligence correlation, or manual inspection. High-risk domains may be temporarily restricted or blocked, depending on asset criticality and organizational policy.

The threshold selection problem was formulated as the minimization of the expected operational cost:

$$C(\tau_1, \tau_2) = c_{FN} \cdot FN(\tau_1, \tau_2) + c_{FP} \cdot FP(\tau_1, \tau_2) + c_V \cdot NV(\tau_1, \tau_2) \tag{8}$$

where $FN(\tau_1, \tau_2)$, $FP(\tau_1, \tau_2)$ and $NV(\tau_1, \tau_2)$ denote the numbers of false negatives, false positives, and domains forwarded for additional verification (elevated-risk category). The parameters c_{FN}, c_{FP}, and c_V represent the costs of a missed threat, a false alarm, and a single verification action, respectively. The cost parameters were selected to represent a plausible industrial monitoring scenario rather than being calibrated against empirical SOC

operational data. The ratio c_{FN}/c_{FP} = 2.5 reflects the assumption that a missed threat, such as undetected command-and-control communication or a phishing attempt targeting an industrial operator, is substantially more costly than a false alarm requiring additional domain verification. The verification cost c_V = 2 reflects the assumption that automated or semi-automated secondary analysis is less costly than either type of misclassification. These values were not calibrated to any specific operational environment. Sensitivity analysis was performed to assess threshold stability under alternative cost configurations, including scenarios with elevated c_{FP} for high-availability production environments and elevated c_{FN} for critical infrastructure.

4. Results

4.1. Exploratory Analysis of Extracted Features

The extracted domain-level features were analysed before model training to identify differences between safe and malicious domains. The analysis covered selected infrastructural, reputation-related, lexical, structural, and geographic attributes, including WHOIS availability, domain age, expiration time, Tranco presence, registrar group, TLD category, and registrant country. Domain age was one of the most informative attributes, with a median of 8945 days for safe domains and -1 for malicious domains, where -1 denoted unavailable WHOIS information retained as a diagnostic value. The distribution of domain age is presented in Figure 4.

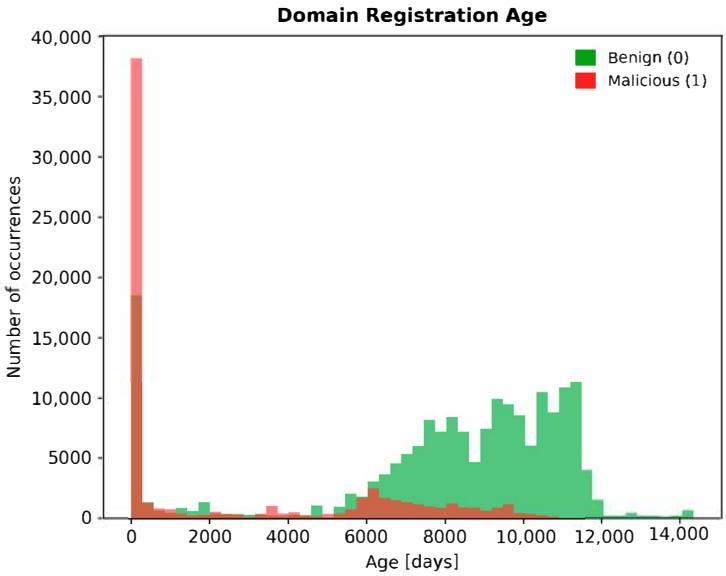


Figure 4. Distribution of domain registration age by class.

A related WHOIS-based attribute was the number of days until domain expiration. Its median value was 277 days for safe domains and -1 for malicious domains, where -1 denoted unavailable WHOIS data. This suggests that registration-related information may be useful not only through domain age, but also through the availability of expiration data. The distribution of this feature is shown in Figure 5.

The distribution of WHOIS registrant countries across the two classes is presented in Figure 6. Country information was unavailable for 66.6% of records and was excluded from the figure. Among domains with known registrant countries, the United States accounted for the largest share and had a malicious rate of 5.1%, while Brazil and China showed rates of 79.9% and 67.5%, respectively. These differences indicate an association between registrant country and class distribution, although the high proportion of missing data

and limited geographic documentation restrict the assessment of representativeness and model generalizability.

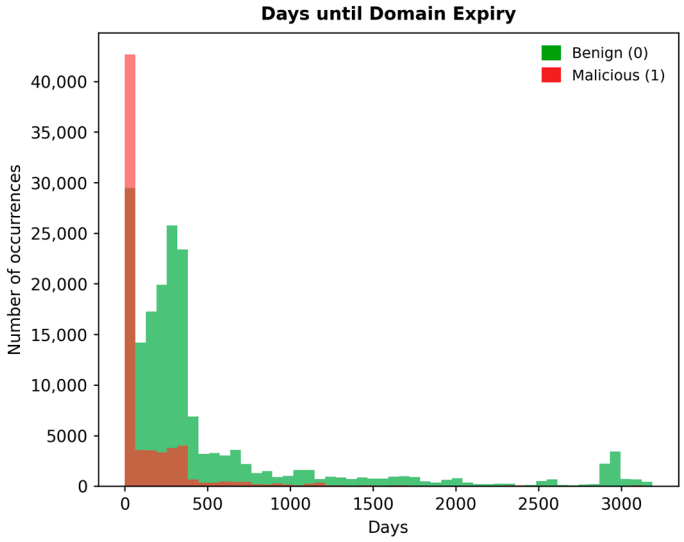


Figure 5. Distribution of days until domain expiration by class.

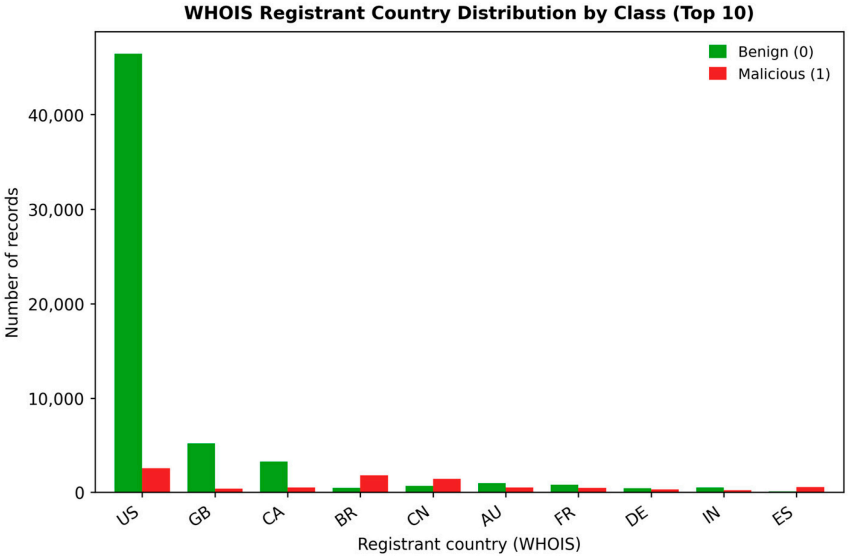


Figure 6. Distribution of WHOIS registrant countries by class.

Unavailable WHOIS information was observed for 56.1% of malicious domains, which indicates that the lack of registration data may support the detection of malicious websites. A similarly clear difference was observed for the Tranco-based reputation feature. In the safe class, 67.8% of domains were present in the Tranco Top 1M ranking, whereas only 4.6% of malicious domains appeared in this ranking. The distribution of this binary feature is shown in Figure 7.

The relationship between registrar group and class label is presented in Figure 8. Domains assigned to the Unknown registrar group were more frequent in the malicious class, whereas domains registered by recognizable corporate registrars were observed mainly in the safe class. This suggests that registrar information complements other WHOIS-based attributes and helps describe differences in domain registration patterns. Figure 9 shows the distribution of TLD categories, where non-standard or less common TLDs occurred more often among malicious domains.

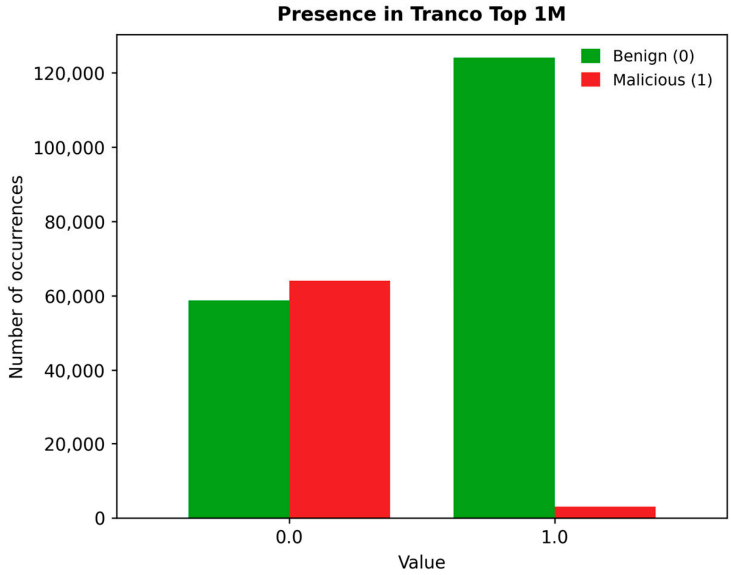


Figure 7. Presence in the Tranco Top 1M ranking by class.

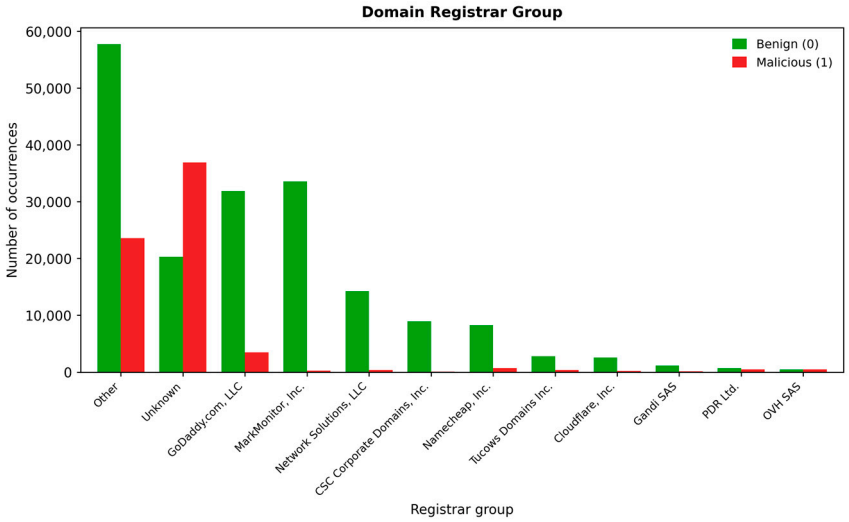


Figure 8. Domain registrar group distribution by class.

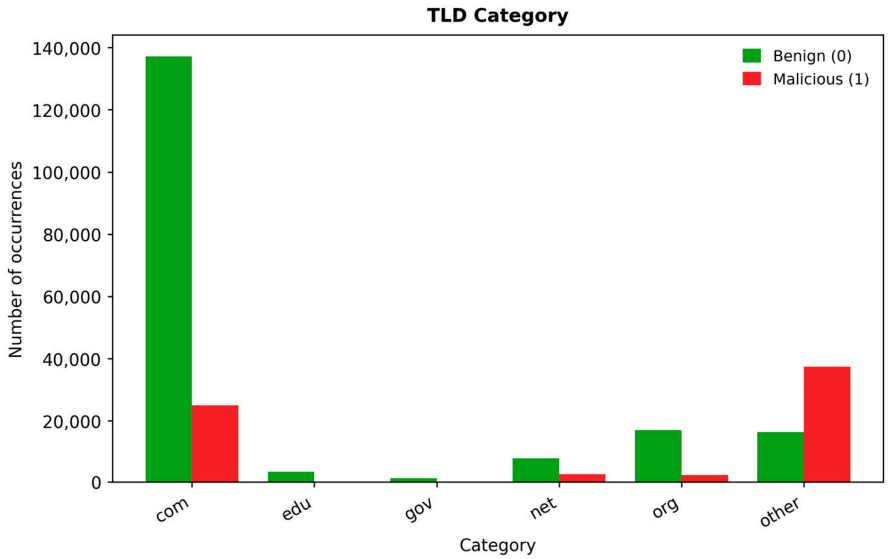


Figure 9. Distribution of top-level domain categories by class.

The exploratory analysis showed differences between safe and malicious domains in selected registration-related, reputation-related and structural attributes. Malicious domains more often lacked accessible WHOIS data, while safe domains were more frequently present in the Tranco ranking. Differences were also observed for registrar group and TLD category. These observations indicate that the analysed attributes may contribute to distinguishing between the two classes in the subsequent classification task.

4.2. Model Performance Comparison

The classification results obtained on the test set using the default binary decision threshold of $\tau = 0.5$ are presented in Table 3. The compared models differed in terms of overall performance and the balance between precision and recall. In malicious domain detection, this balance is important because high recall reduces the number of missed malicious domains, while high precision limits the number of false alarms.

Table 3. Model performance on the test set (n = 59,508).

Model	ROC AUC	PR AUC	F1	MCC	Accuracy	Precision	Recall
XGBoost	0.9741	0.9372	0.8350	0.7764	0.9076	0.7724	0.9088
Random Forest	0.9683	0.9222	0.8325	0.7771	0.9161	0.8561	0.8101
AdaBoost	0.9611	0.9051	0.7942	0.7221	0.8780	0.7019	0.9143
SVM	0.9561	0.8944	0.8064	0.7362	0.8934	0.7575	0.8620
Logistic Regression	0.9512	0.8822	0.7748	0.6956	0.8644	0.6766	0.9064

XGBoost achieved the highest ROC AUC of 0.9741, PR AUC of 0.9372 and F1-score of 0.8350. It also maintained high recall equal to 0.9088, with precision of 0.7724 and accuracy of 0.9076. Random Forest obtained the highest MCC of 0.7771, accuracy of 0.9161 and precision of 0.8561, but its recall was lower at 0.8101. AdaBoost and Logistic Regression reached recall values above 0.90, although their lower precision indicates a higher number of false alarms. These results show that XGBoost provided the strongest overall performance, while Random Forest was more conservative in assigning the malicious class.

In addition to predictive performance, the computational and operational characteristics of the proposed method were evaluated. The analysis covered model training time and the processing time of individual pipeline modules. These measures were used to assess the suitability of the method for periodic retraining and repeated domain assessment in cybersecurity monitoring workflows. The training times of the five evaluated classifiers are summarized in Table 4.

Table 4. Training time of the compared classification models.

Model	Training Time (s)
XGBoost	57
Random Forest	190
AdaBoost	2199
SVM	37,644
Logistic Regression	1889

XGBoost had the shortest training time at 57 s, followed by Random Forest at 190 s, Logistic Regression at 1889 s, and AdaBoost at 2199 s. SVM was the most computationally

demanding model, requiring 37,644 s, equivalent to approximately 10.5 h. Considering predictive performance and training cost, XGBoost was selected for further qualitative and external validation, while the long training time of SVM limits its applicability in workflows requiring periodic model updates. Parallel execution can reduce training time, although the achieved speedup depends on the classifier and parallelization strategy [21]. The high computational cost observed for SVM reflects the use of an RBF kernel, whose training complexity increases superlinearly with the number of samples [16], while linear formulations and kernel approximation methods offer greater computational efficiency.

The processing latency of the selected XGBoost-based pipeline was evaluated using the 31 domains included in the qualitative verification sample. Each domain was processed five times, and wall-clock time was recorded separately for the individual feature groups, model inference, and complete end-to-end assessment. Network-dependent modules used connection and read timeouts of 6 s and 12 s, respectively. This setup allowed the latency contribution of local computation, external data acquisition, and model inference to be compared under the same measurement conditions. The median value calculated from the repeated measurements is reported for each module in Table 5.

Table 5. Module-level processing time of the domain assessment pipeline.

Module	Median Time (ms)
Lexical, structural, and Levenshtein-based features	0.4
Tranco-based features	<0.1
DNS features	709.3
SPF, DKIM, and DMARC features	997.4
WHOIS features	1141.3
HTTP features	1208.7
XGBoost prediction	235.9
End-to-end processing	4213.6

Locally computed lexical, structural, Levenshtein-based, and Tranco-related features introduced negligible processing overhead. Network-dependent modules accounted for most of the end-to-end latency, with the highest median times recorded for HTTP and WHOIS data acquisition. Prediction using the selected XGBoost model required 235.9 ms, while the independently measured median end-to-end processing time was 4213.6 ms per domain. The operational performance of the pipeline was therefore determined primarily by external data acquisition rather than by model inference. Parallel queries, asynchronous processing, caching, and periodic updating of network-based attributes may reduce processing latency in practical monitoring systems.

4.3. Discriminative Power of Feature Groups

To interpret the contribution of individual variables, feature importance was assessed using the ANOVA F-test on the training set. The eight highest-ranked predictors are presented in Table 6. The two strongest attributes were domain_age_days and tranco_in_top1m, with F-statistics of 86,344 and 86,302, respectively. Their values were approximately three times higher than the eighth-ranked feature, tranco_in_top10k, whose F-statistic was 28,458.

Table 6. Feature ranking according to the ANOVA F-statistic.

Rank	Feature	Category	*F*
1	domain_age_days	Infrastructural	86,344
2	tranco_in_top1m	Reputation-related	86,302
3	domain_type_other	Infrastructural	68,790
4	whois_registered	Infrastructural	57,845
5	whois_registrar_group_Unknown	Infrastructural	51,307
6	url_param_count	Structural	40,873
7	dns_resolves	Infrastructural	28,508
8	tranco_in_top10k	Reputation-related	28,458

The ranking indicates that the model relied mainly on infrastructural and reputation-related information. WHOIS variables, domain age, registrar group, and Tranco presence showed stronger discriminatory power than most lexical and URL-structure attributes. Among the eight highest-ranked predictors, only url_param_count was structural, indicating that URL construction provided additional but limited information.

To validate the ANOVA-based ranking, feature importance was additionally assessed using SHAP values with TreeExplainer on a 5000-sample test subset, XGBoost gain-based importance, and permutation importance on the full test set using ROC AUC over 10 repetitions. The cross-method comparison of features appearing in the top 15 of at least three methods is presented in Table 7.

Table 7. Cross-method comparison of feature importance rankings.

Feature	Category	ANOVA	SHAP	XGBoost Gain	Permutation
domain_age_days	Infrastructural	1	4	15	1
dns_resolves	Infrastructural	7	2	1	2
days_to_expire	Infrastructural	15	3	11	3
dns_total_records	Infrastructural	11	7	3	4
dns_a_count	Infrastructural	10	12	12	6
domain_type_other	Infrastructural	3	9	2	8
url_param_count	Structural	6	1	–	–
tranco_in_top1m	Reputation	2	–	–	–

Note: "–" indicates that the feature was not ranked in the top 15 by the respective method.

Six features appeared in the top 15 across all four methods, five of which belong to the infrastructural category. This cross-method consistency supports the conclusion that domain lifecycle and DNS infrastructure characteristics are among the primary discriminative signals across the evaluated importance methods. Minor discrepancies were observed for lexical and reputation-related features. url_param_count ranked first by SHAP but was absent from the top 15 for Gain and Permutation, while tranco_in_top1m ranked second by ANOVA but did not appear among the top 15 for the model-based methods. These differences indicate that the importance of some features depends on the evaluation method and may be partly captured by stronger infrastructural predictors.

4.4. Error Analysis and Temporal Validity

A systematic analysis of classification errors was performed on the test subset of the malicious_phish dataset to identify domain characteristics associated with false positives (n = 4103) and false negatives (n = 1397) at the default decision threshold of $\tau = 0.5$. The results, grouped by availability, WHOIS registration, domain age, registrar, TLD category, DNS infrastructure, and reputation, are presented in Table 8.

Table 8. Feature profiles of false-positive and false-negative classifications on the test set.

Feature	*FP* (n = 4103)	*FN* (n = 1397)
DNS data availability	52.7%	80.7%
WHOIS data availability	53.3%	93.2%
Median domain age	5421 days	6572 days
Unknown registrar	45.3%	6.5%
Share of .com domains	45.9%	70.8%
Share of other TLDs	25.3%	8.9%

False positives were often associated with sparse or incomplete infrastructure. Only 52.7% resolved via DNS and 53.3% had valid WHOIS data, while 45.3% had an unknown registrar. False negatives showed a more complete infrastructure profile, with 80.7% resolving via DNS, 93.2% having valid WHOIS data, and 70.8% associated with .com domains. This suggests that false negatives were more difficult to distinguish from domains classified as benign using the proposed feature set.

The temporal gap between the class labels and extracted features remains an important limitation of the internal evaluation. The labels originate from the malicious_phish dataset published in 2021, whereas the feature values describe the domains in April 2026. Consequently, attributes such as whois_registered, dns_resolves, and the −1 value of domain_age_days may partly reflect domain lifecycle status rather than maliciousness itself. This interpretation is consistent with the prominence of these attributes in the feature ranking and with the profile of missed PhishDestroy domains, for which WHOIS data were almost always available. Temporal effects of this type are a known source of experimental bias in security-oriented machine learning [22], while WHOIS availability may also be affected by privacy redaction introduced after the GDPR [23].

To quantify this effect, XGBoost was evaluated on the subset of test-set domains that remained resolvable and had valid WHOIS records at the time of feature extraction in April 2026. The subset was defined by dns_resolves = 1 and whois_registered = 1 and comprised 42,455 of the 59,508 test domains (71.3%), including 6053 malicious domains (14.3%), compared with 25.7% in the full test set. This confirms that expired or unavailable malicious domains constitute a substantial share of the historical test set. The results are presented in Table 9.

Table 9. Comparison of model performance for the full test set and active-domain subset.

Subset	n	Malicious (%)	ROC AUC	PR AUC	F1	Recall
Full test set	59,508	25.7%	0.9741	0.9372	0.8350	0.9088
Active domains (dns = 1, whois = 1)	42,455	14.3%	0.9716	0.8824	0.7655	0.8209

The ROC AUC decreased only marginally from 0.9741 to 0.9716 (−0.003), whereas PR AUC decreased from 0.9372 to 0.8824 (−0.055), F1-score from 0.8350 to 0.7655 (−0.070), and

recall from 0.9088 to 0.8209 (−0.088). The near-invariance of ROC AUC indicates that the model retained similar ranking performance for active domains, while the larger changes in PR AUC, F1, and recall reflect the more difficult classification setting. The active-domain subset also contains a lower proportion of malicious domains (14.3% vs. 25.7%), which affects threshold-dependent metrics independently of model quality. It therefore provides a more conservative assessment of performance for operationally active domains. The full-test-set ROC AUC of 0.9741 should consequently be interpreted with caution, while the external evaluation on 1000 active PhishDestroy domains (recall = 70%) provides an additional benchmark for operational effectiveness.

To further assess whether the observed performance was affected by the representation of unavailable WHOIS data, an ablation study was conducted using the feature preprocessing described in Section 3.3. Variant A corresponds to the baseline model, with −1 retained as a sentinel value. In Variant B, −1 values were replaced with the training-set median (8971 days and 298 days, respectively), while the existing binary availability flags (dns_resolves, whois_registered, whois_age_known) were retained. In Variant C, median imputation was applied and these availability flags were removed. The results are presented in Table 10.

Table 10. Ablation study of missing-value encoding on the test set.

Variant	ROC AUC	PR AUC	F1	Precision	Recall
A—Baseline (−1 sentinel)	0.9741	0.9372	0.8350	0.7724	0.9088
B—Median imputation, flags retained	0.9750	0.9392	0.8305	0.7587	0.9172
C—Median imputation, flags removed	0.9751	0.9395	0.8355	0.7662	0.9185

All three variants produced comparable results, with differences in ROC AUC below 0.001 and in F1-score below 0.005. Variants B and C achieved marginally higher AUC values than the baseline, indicating that performance did not depend on the −1 representation alone. The similar results for Variants B and C also suggest that the availability flags provided limited additional information after median imputation. Overall, the encoding scheme did not materially affect model performance, indicating that the observed predictive signal was not dependent on the −1 representation alone.

4.5. External Validation of the Selected Model

The selected XGBoost model was then evaluated on external domains that were not included in the training dataset. The validation was performed in two stages. First, 31 manually selected domains were analyzed to verify whether the model scores were consistent with interpretable domain-level characteristics. Second, the model was tested on 1000 active phishing domains from PhishDestroy [24], which allowed for assessing detection performance under more realistic conditions.

Selected examples from the qualitative verification are presented in Table 11. Malicious domains classified with high probability were usually characterized by unavailable WHOIS data, uncommon TLDs, lack of Tranco ranking presence or short registration history. In contrast, domains classified as safe with low maliciousness probability were present in the Tranco Top 10K ranking and had long registration histories.

The analysis of individual cases revealed two important limitations of the model. A false positive was observed for eduroam.pl, a legitimate academic domain whose administrative and ranking profile differs from globally popular commercial domains, mainly due

to a national TLD and a registrar pattern underrepresented in the training data. For similar reasons, gov.pl, although correctly classified as benign at the default binary threshold, obtained an elevated score of 43.53% and fell into the elevated-risk zone of the three-tier scale. A false negative was observed for a phishing domain whose infrastructure resembled a legitimate service, indicating that domain-level infrastructure and reputation features may be insufficient when attackers use more credible registration characteristics.

Table 11. Selected cases from qualitative verification on external domains.

Domain	Maliciousness Score	Result
refriplus.com.br	98.89%	*TP*
landingcaptcha12.live	93.13%	*TP*
aichaingpt.top	81.55%	*TP*
cnn.com	1.19%	*TN*
paypal.com	2.92%	*TN*
eduroam.pl	64.18%	*FP*
gov.pl	43.53%	*TN*
paypal-secure-login.com	21.31%	*FN*

The second validation stage was performed on 1000 active phishing domains from PhishDestroy. At the default decision threshold of 0.5, the model detected 700 domains, giving a recall of 70%. This result was lower than the recall obtained on the historical test set, which indicates that current phishing domains may have feature profiles different from those represented in the training data. To identify the source of this difference, detected and missed domains were compared using selected registration and TLD-related attributes. The comparison is presented in Table 12.

Table 12. Feature profile of PhishDestroy domains (n = 1000).

Feature	Detected (n = 700)	Missed (n = 300)
WHOIS data availability	26.1%	95.3%
Median domain age	−1 days	226 days
Share of .com domains	15.1%	66.0%
Share of non-standard TLDs	84.3%	21.7%

The detected phishing domains were mainly characterized by unavailable WHOIS data, non-standard TLDs and lack of registration history. The missed domains had a different profile. WHOIS data were available in 95.3% of these cases, the median domain age was 226 days, and 66.0% of the missed domains used the .com TLD. This shows that active phishing domains with complete registration data and common TLDs are more difficult to detect using the proposed domain-level representation.

4.6. Cost-Based Threshold Optimization

The cost-based threshold optimization described in Section 3.7 was performed using out-of-fold prediction scores generated from the training subset through five-fold grouped cross-validation. The base domain was used as the grouping variable to prevent URLs associated with the same domain from appearing in both the training and validation portions of a fold. The out-of-fold scores from all validation folds were combined, and the

decision thresholds were selected by minimizing the assumed operational cost. The test set and the external PhishDestroy sample were not used during threshold selection.

For the binary setting, the threshold was optimized over discrete candidate values $\tau \in \{0.05, 0.10, \ldots, 0.95\}$. Under the base cost configuration $c_{FN} = 10$ and $c_{FP} = 1$, the optimal threshold was $\tau^* = 0.50$. This suggests that, for the assumed cost ratio, the default threshold remained a suitable decision boundary for binary classification.

For the three-tier decision framework, pairs of thresholds (τ_1, τ_2) were evaluated. Under the cost configuration $c_{FN} = 10$, $c_{FP} = 4$, and $c_V = 2$, the optimization yielded $\tau_1^* = 0.30$ and $\tau_2^* = 0.70$. This configuration reflects a scenario in which a missed threat is substantially more costly than a false alarm, while additional verification is less costly than incorrectly blocking a legitimate domain. The sensitivity of the optimal upper threshold τ_2^* to increasing c_{FP}, with $c_{FN} = 10$ and $c_V = 2$, is presented in Table 13.

Table 13. Optimal upper threshold τ_2^* as a function of c_{FP} ($c_{FN} = 10$, $c_V = 2$).

c_{FP}	τ_1^*	τ_2^*	$C(\tau_1^*, \tau_2^*)$
1.0	0.30	0.50	17,667
3.0	0.30	0.55	25,698
3.5	0.30	0.65	27,028
4.0	0.30	0.70	27,992
4.5	0.30	0.75	28,871
5.0	0.30	0.80	29,487

The results show that the lower threshold τ_1^* remained stable at 0.30, while the upper threshold τ_2^* increased as the cost of false positives became higher. This indicates that the width of the elevated-risk zone depends mainly on the assumed cost of false alarms relative to additional verification. To assess threshold behaviour across a broader range of industrial deployment scenarios, the optimization was repeated for six cost configurations. The results are presented in Table 14.

Table 14. Cost-based threshold optimization under different operational scenarios.

Scenario	c_{FN}	c_{FP}	c_V	τ_1^*	τ_2^*	Recall	FPR
Baseline	10	4	2	0.30	0.70	0.830	0.043
High FP cost	10	5	2	0.30	0.80	0.751	0.022
Very high FP cost	10	8	2	0.30	0.85	0.706	0.014
High FN cost	20	4	2	0.20	0.70	0.830	0.043
High FN and FP costs	20	8	2	0.20	0.85	0.706	0.014
High verification cost	10	4	5	0.45	0.50	0.909	0.093

The lower threshold $\tau_1^* = 0.30$ remained stable at $c_{FN} = 10$ and decreased to 0.20 when c_{FN} doubled to 20. The upper threshold τ_2^* increased from 0.70 to 0.85 as c_{FP} increased, reducing FPR from 4.3% to 1.4% while recall decreased from 83.0% to 70.6%. When c_V increased to 5, τ_1^* increased to 0.45 and τ_2^* decreased to 0.50, narrowing the elevated-risk range. These results show that threshold selection depends on the assumed operational costs and should be calibrated to the deployment context.

Using the cost-optimized thresholds, the three-level risk scale was applied to the PhishDestroy sample, with low risk below 30%, elevated risk from 30% to 70%, and high

risk at or above 70%. Of the 1000 active phishing domains, 500 were assigned to the high-risk zone, 349 to the elevated-risk zone, and 151 to the low-risk zone. Thus, 84.9% were assigned to either the high- or elevated-risk zone. The precision, recall, and false-positive rate for the same risk zones on the malicious_phish test subset are presented in Table 15.

Table 15. Classification performance across the three risk zones on the test set.

Risk Zone	Threshold	*n*	% Total	Malicious	Benign	Precision	Recall *	FPR †
Low risk	$p < 0.30$	38,472	64.7%	754	37,718	2.0%	4.9%	85.4%
Elevated	$0.30 \leq p < 0.70$	6434	10.8%	1858	4576	28.9%	12.1%	10.4%
High risk	$p \geq 0.70$	14,602	24.5%	12,706	1896	87.0%	82.9%	4.3%

Note: * Share of all malicious domains in the test set assigned to this zone, † Share of all benign domains in the test set assigned to this zone.

The high-risk zone captured 82.9% of all malicious domains with a precision of 87.0%, while only 4.3% of benign domains were assigned to this zone. The low-risk zone covered 64.7% of all domains and contained 2.0% malicious records. The 754 malicious domains assigned to the low-risk zone represented 4.9% of all malicious domains. The elevated zone covered 10.8% of domains and had a malicious rate of 28.9%, making it the primary zone for additional verification. The distribution of PhishDestroy domains across the three risk zones is shown in Figure 10.

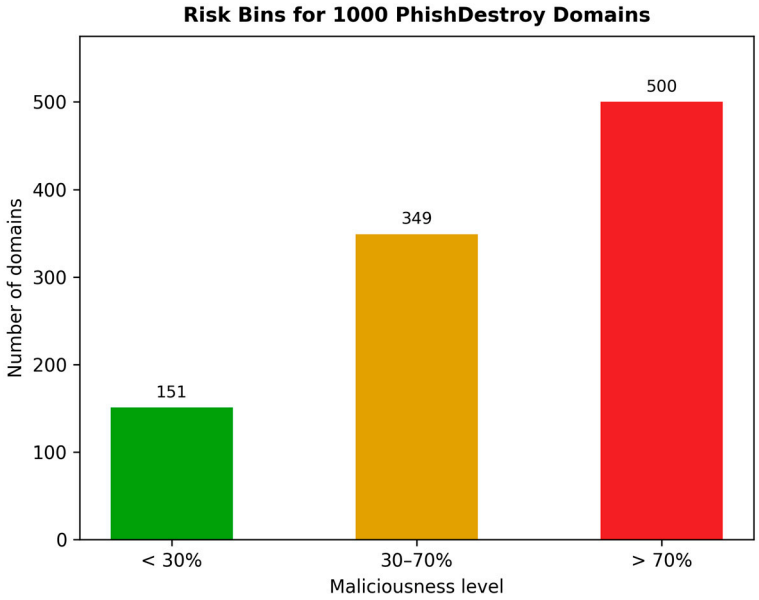


Figure 10. Distribution of PhishDestroy domain scores across three risk zones.

The validation results indicate that the model generalized well to phishing domains with incomplete registration data, uncommon TLDs, and no reputation signals. However, its effectiveness decreased for domains whose infrastructure resembled legitimate services, especially when WHOIS data were available and common TLDs were used. This suggests that a single binary threshold may be insufficient in practical deployment, where missed threats and excessive false alarms may both affect security monitoring. Therefore, the model may be more useful as a risk scoring component than as a standalone binary classifier. In cybersecurity monitoring pipelines, such scores can support the prioritization of suspicious domains, while domains with intermediate scores can be forwarded for additional verification instead of being classified directly as safe.

5. Conclusions

The results of the conducted study indicate that, among the five compared classification algorithms, XGBoost achieved the most favorable overall performance in malicious domain detection. The model obtained the highest ROC AUC and PR AUC values and provided a good balance between recall and general classification quality. Feature importance analysis using the ANOVA F-test, SHAP values, XGBoost gain, and permutation importance consistently identified domain age, DNS infrastructure attributes, and Tranco presence as the strongest predictors, whereas URL-structure attributes had lower discriminatory power across all evaluated methods.

External validation on active PhishDestroy domains showed that high performance on a historical test set does not guarantee the same effectiveness against current phishing campaigns. At the default threshold, the selected model achieved a recall of 70%. After introducing a three-level risk scale, 84.9% of active phishing domains were classified as malicious or suspicious. This indicates that the method is better suited as a risk scoring and decision-support component than as a purely binary filter.

The practical value of the study is that domain-level models can detect many malicious websites without downloading page content. In cybersecurity monitoring pipelines, such models may provide an additional data-driven signal for prioritizing suspicious domains and supporting protective decisions. This can be relevant for organizational and industrial environments, where missed threats and excessive false alarms may both affect security operations. The main limitation is the extraction of network-based features, especially DNS, WHOIS, and HTTP attributes. The results also indicate that phishing domains with complete WHOIS data, common TLDs, and infrastructure resembling legitimate services remain more difficult to detect.

Further research should focus on extending the proposed method with temporal features describing changes in domain infrastructure, such as DNS record variability, registrar changes and short-term HTTP availability. Another direction is uncertainty-aware classification, in which domains near the decision boundary are forwarded for additional verification instead of being assigned directly to one class. Future work should also include aging experiments to assess how quickly individual feature groups lose diagnostic value over time.

From the perspective of industrial systems, future work should address the integration of domain-level risk scoring with OT network monitoring. This includes evaluation on DNS traffic collected at the IT/OT boundary, deployment within industrial demilitarized zones, correlation of domain risk scores with SCADA and IIoT events in SIEM platforms, and analysis of feature-acquisition latency and caching strategies. Such studies would enable validation under the operational constraints of industrial control systems.

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